

VIKAS SHARMA

Software Development Engineer | Distributed Systems | High-Throughput Platforms | Python & C++

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PROFESSIONAL SUMMARY

Software Engineer with 3+ years designing and building production-grade distributed systems, SDKs, and scalable, high-throughput data platforms. Architected infrastructure handling **\$48M+ in daily transaction volume across 150+ connected clients**, engineered **150x performance improvements** through vectorized computation and parallel orchestration, and published **pyzdata** on PyPI (Production/Stable). Strong in system design, distributed messaging, concurrency, fault tolerance, performance engineering, and CI/CD.

TECHNICAL SKILLS

Languages: Python (Expert; object-oriented design, type hints), C++, SQL, Bash

Distributed Systems: ZeroMQ (message-passing), REST APIs, WebSocket, microservices, state machines, crash recovery, idempotent restarts

Concurrency & Perf: AsyncIO, Multiprocessing, Multithreading, FileLock orchestration, NumPy vectorization, O(log n) lookup structures

Data Engineering: NumPy, Polars, Pandas, DuckDB, Parquet, ETL pipelines, tick-level time-series processing, in-memory analytics

Cloud & DevOps: AWS EC2, Docker, Linux, Git, GitHub Actions, CI/CD, trusted publishing, Sigstore-signed releases

Web & Frameworks: Django, Flask, FastAPI, Streamlit, REST API design, pytest, Pydantic

PROFESSIONAL EXPERIENCE

Software Engineer II - Pranjali Growcap Pvt. Ltd.

Feb 2024 - Present

Proprietary fintech firm - high-throughput transaction processing

New Delhi, India

- Distributed Transaction Platform:** Architected a production system serving **150+ connected clients with \$48M in daily transaction volume** - real-time validation, reconciliation, and end-to-end audit trails across 3 third-party service integrations; eliminated 100% of manual workflows.
- Microservice Architecture:** Designed a **ZeroMQ hub-spoke system** - central message broker coordinating 5 independent worker processes via asynchronous message-passing; CSV-backed state machines for crash recovery and idempotent restarts; fully autonomous on AWS EC2 with zero manual intervention.
- Performance Engineering (150x speedup):** Optimized a data-intensive batch processing engine from **~150 hours to under 1 hour** through NumPy vectorization, O(log n) binary-search lookups, and 16-way parallel orchestration via FileLock-based job distribution over shared storage.
- High-Volume Data Pipelines:** Built ETL systems processing 8+ years of timestamped tick-level time-series datasets into columnar Parquet storage; served through a Polars + DuckDB analytics layer exposed via Streamlit dashboards - **50,000+ job combinations processed per cycle**.
- Fault Tolerance:** Engineered crash-recovery subsystem using CSV-based state machines and file-backed idempotency tokens, enabling automated restarts without duplication or state loss across all downstream services.

Software Engineer - Pranjali Growcap Pvt. Ltd.

Jul 2022 - Jan 2024

Proprietary fintech firm

New Delhi, India

- Built v1 of the firm's internal platform SDK integrating multiple third-party REST and WebSocket APIs with real-time data streaming, session management, and automated retry logic.
- Developed a reproducible in-house data-analysis framework replacing ad-hoc spreadsheet workflows across internal teams.
- Shipped monitoring and alerting stack (Streamlit + Telegram bots) for real-time operational dashboards and system-health checks.
- Designed ETL pipelines for cleaning and structuring high-volume upstream data feeds.

PROJECTS

PyZData (pip install pyzdata) - Published on PyPI | *v1.0.6, Production/Stable, MIT License*

- Python package + CLI + Streamlit web app for historical financial data. Package engineering includes **90 unit tests, PEP 561 typing, trusted publishing via GitHub Actions (Sigstore-signed releases), semantic versioning, and Python 3.10-3.13 support**. **32 GitHub stars, 15 forks**.

KiteWeb (formerly APICostFree-Zerodha-Trade) (28 stars, 9 forks) - REST wrapper library implementing authentication, session management, and high-level order/market-data endpoints; adopted across the Indian retail algorithmic-trading community.

Developer Productivity Tools - Jupyter context-menu launcher (Windows shell integration), PickleParquetReader (file-association handler), TerminalSequence (multi-monitor terminal grid), Tele_Easy_Bot.

EDUCATION

BSc Computer Science - Shaheed Sukhdev College of Business Studies, University of Delhi

2021 - 2023

CODING PROFILES

Data structures and algorithms practice: **CodeChef:** 2-star (rating 1495, peak 1559), 252 problems solved, 10 rated contests

LeetCode: 65 solved (43 easy, 21 medium, 1 hard) **HackerRank:** SQL Intermediate + 3 verified certs